

# Architecture dependence in the many-body entanglement structure of local quantum circuits

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# INTRODUCTION

Entanglement is hard

How to optimally detect  
entanglement theoretically  
and in the laboratory?

How to characterize,  
control, and quantify  
entanglement?



# INTRODUCTION

**Entanglement is hard... but**

Many interacting systems  Two interacting systems (local)

**Motivation for two-qubits gates.**



- ☐ Minimal model for entanglement.
- ☐ 2-qubit gates have experimental implementation.
- ☐ Universal gate set = Turing complete.

# ENTANGLEMENT

## A bipartite quantum system

$$\hookrightarrow |\psi\rangle = \sum_{i_A, i_B} c_{i_A, i_B} |i_A\rangle \otimes |i_B\rangle \in \mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$$

A bipartite system is entangled  
if it cannot be written as

$$\longrightarrow |\psi\rangle \neq |\psi_A\rangle \otimes |\psi_B\rangle$$

And for mixed states:  $\boxed{\rho \neq \rho_A \otimes \rho_B}$

# ENTROPY AS A METRIC TO MEASURE ENTANGLEMENT

In classical information theory, exist the Shannon entropy.

$$H(X) := - \sum_{x \in \mathcal{X}} p(x) \log p(x)$$



“Number of questions I must ask to find particular information”.

Similarly, for quantum systems the Von Neumann entropy is defined

$$S = -\text{Tr}(\rho \log \rho)$$

**Entropy is used to quantify entanglement**

$$\left\{ \begin{array}{ll} S = 0 & \text{Separable} \\ S = 1 & \text{Maximally entanglement} \end{array} \right.$$

# ENTROPY AS A METRIC TO MEASURE ENTANGLEMENT

📦 Rényi entropy generalizes a set of entropies: Calculate tool

$$S_n = \frac{1}{1-n} \log (\text{Tr}(\rho_A^n)) = \frac{1}{1-n} \log \left( \sum_i \lambda_i^n \right)$$

➡  $\rho_A$  is the reduced matrix of system A, and  $\lambda_i$  are eigenvalues of  $\rho_A$

$$\square \quad n = 0 \quad : \quad S_0 = \log (\text{rank}(\rho_A))$$

$$\square \quad n \rightarrow \infty \quad : \quad S_\infty = \log \left( \frac{1}{\lambda_{\max}} \right)$$

$$\square \quad n \rightarrow 1 \quad : \quad S_1 = S$$

# ENTROPY AS A METRIC TO MEASURE ENTANGLEMENT

## Rényi entropy

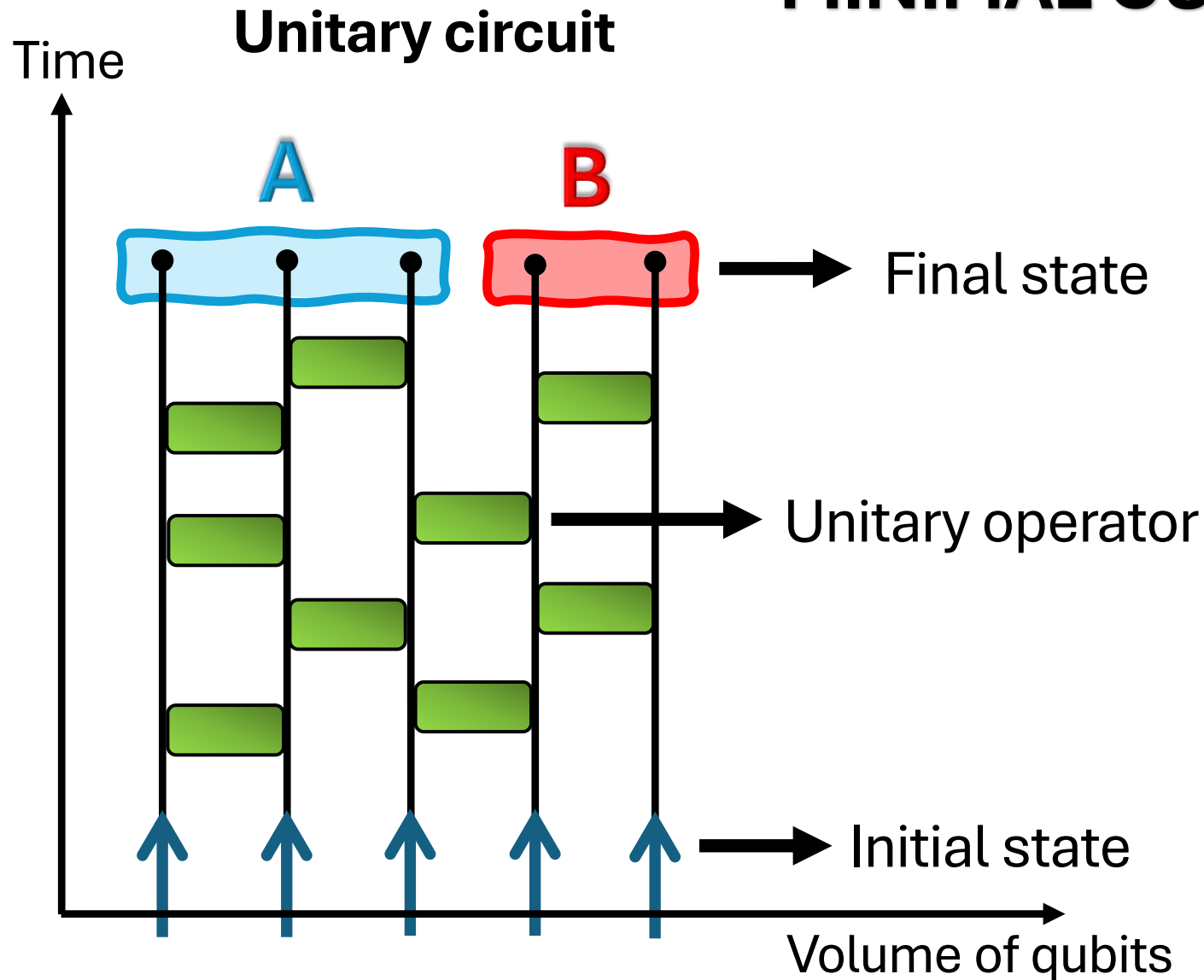
$$S_n = \frac{1}{1-n} \log (\text{Tr}(\rho_A^n)) = \frac{1}{1-n} \log \left( \sum_i \lambda_i^n \right)$$

$S_n$  is bounded

$$\boxed{S_\infty \leq S_n \leq S_0} \quad n \geq 0$$

$$\boxed{S_n \leq \log (\# \text{ of nonzero terms in any decomposition})}$$

# MINIMAL CUT



The quantum state evolves through the application of discrete operators.



Alice

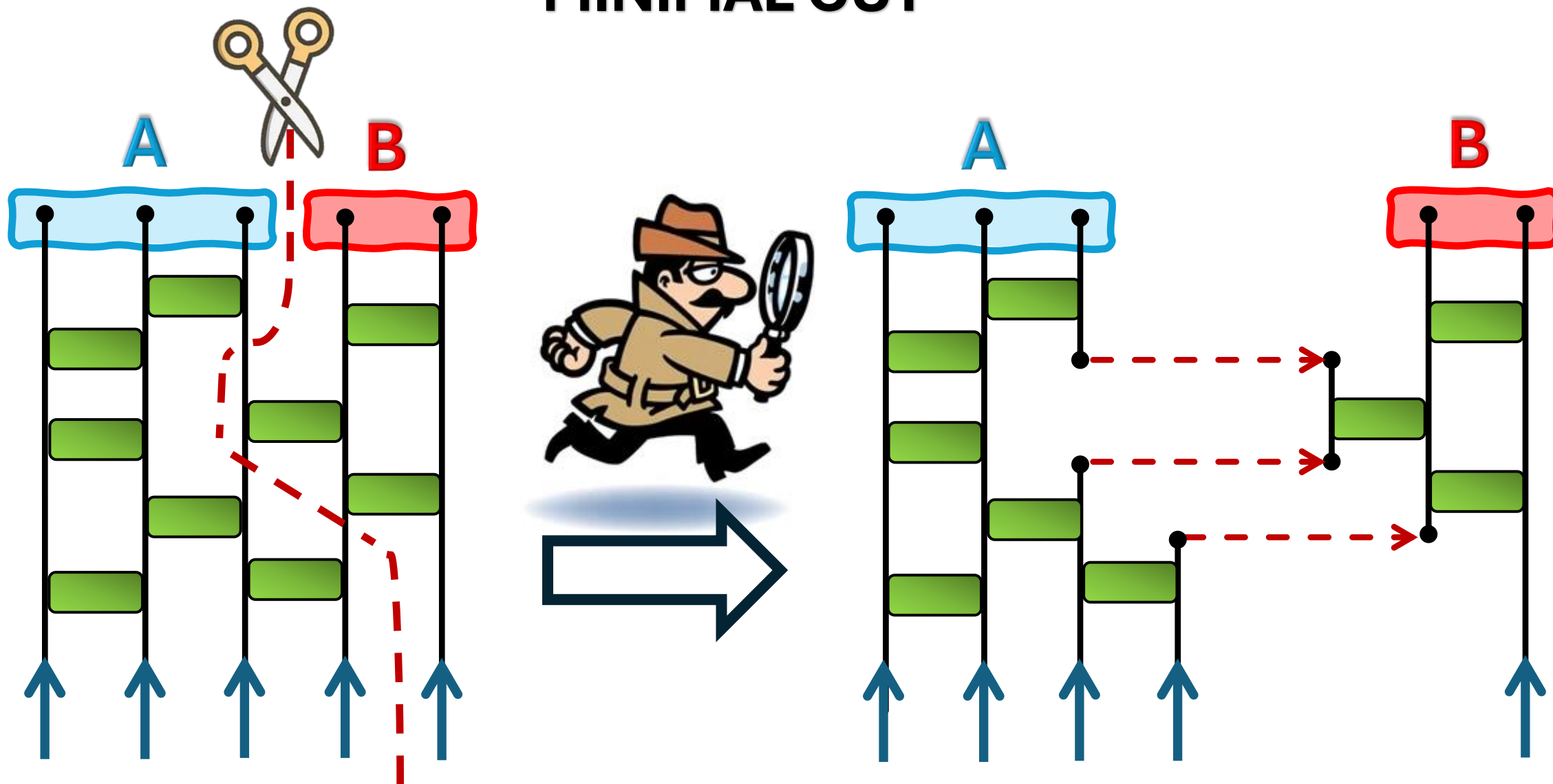
and



Bob

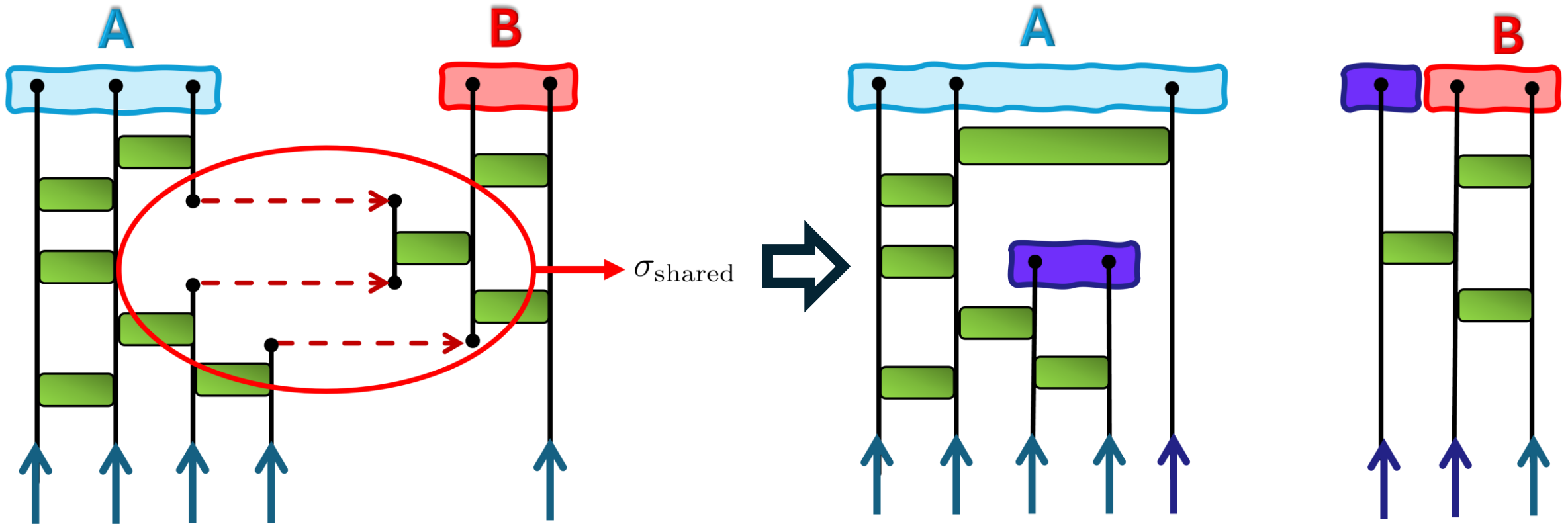
How much entanglement?

# MINIMAL CUT

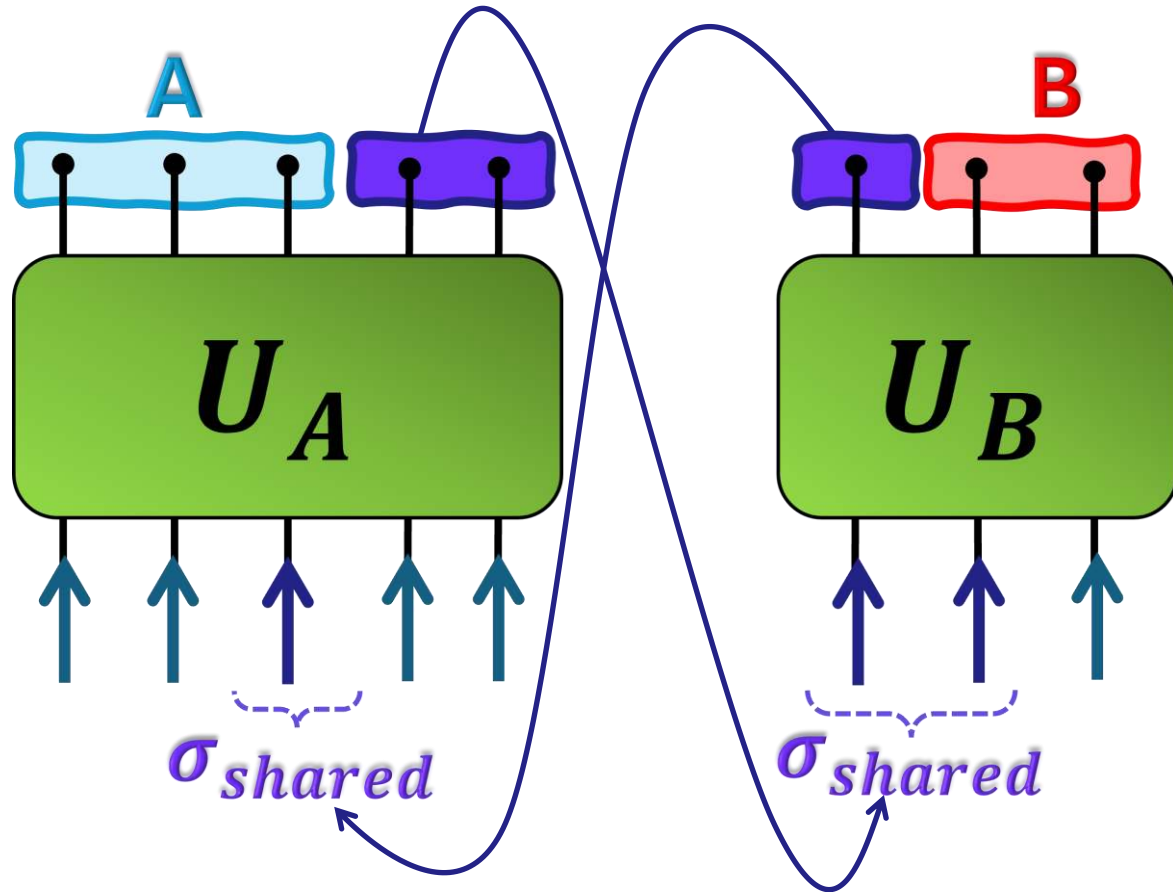


# MINIMAL CUT

$$|\psi\rangle = \sum_{A,B} \sum_{\sigma_{\text{shared}}} a(A, \sigma_{\text{shared}}) |A\rangle \otimes b(B, \sigma_{\text{shared}}) |B\rangle$$



# MINIMAL CUT



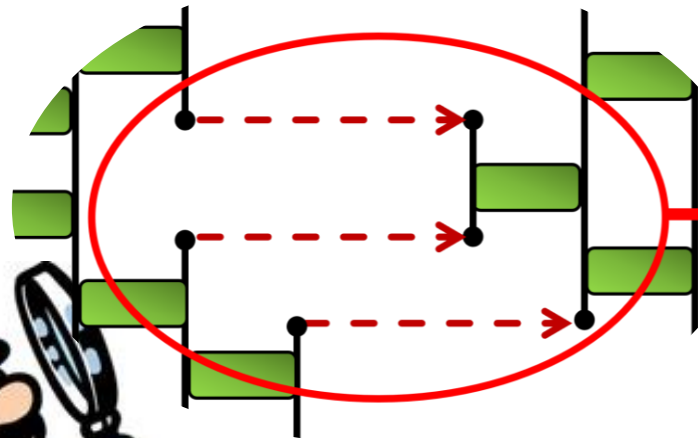
$$\begin{aligned} a(A, \sigma_{shared}) &= \langle A, \sigma_{shared} | U_A | \uparrow \uparrow \uparrow \uparrow \rangle \\ b(B, \sigma_{shared}) &= \langle B, \sigma_{shared} | U_B | \uparrow \rangle \end{aligned}$$

$$|\psi\rangle = \sum_{\sigma_{Shared}} \left( \sum_A a(A, \sigma_{Shared}) |A\rangle \right) \otimes \left( \sum_B b(B, \sigma_{Shared}) |B\rangle \right)$$

# MINIMAL CUT

$$|\psi\rangle = \sum_{\sigma_{\text{Shared}}} \left( \sum_A a(A, \sigma_{\text{Shared}}) |A\rangle \right) \otimes \left( \sum_B b(B, \sigma_{\text{Shared}}) |B\rangle \right)$$

Number of terms in this decomposition



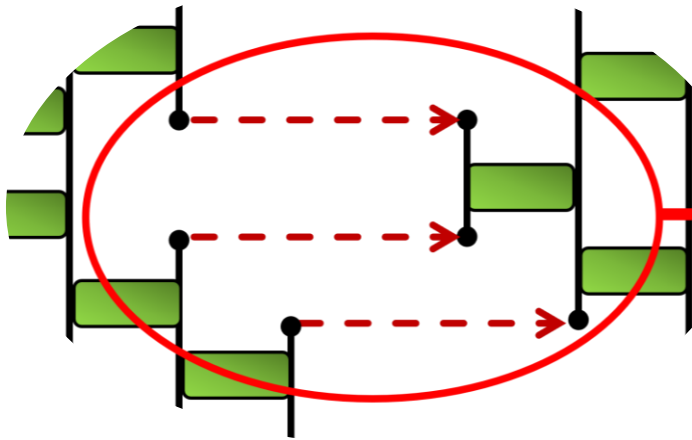
Number of cuts =  $N_{\text{cut}}$

....and remembering  $S_n \leq \log(2^{N_{\text{cut}}})$

$$S_n \leq N_{\text{cut}}$$



# MINIMAL CUT



Number of minimal cut =  $N_{\text{min cut}}$

The tightest **classical** upper bound

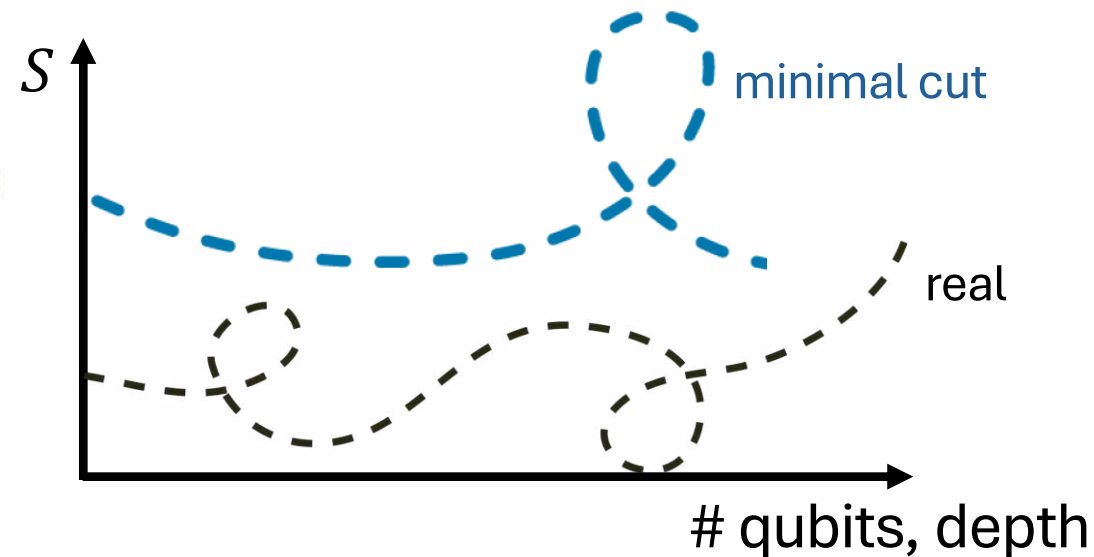
$$S_n \leq N_{\text{min cut}}$$

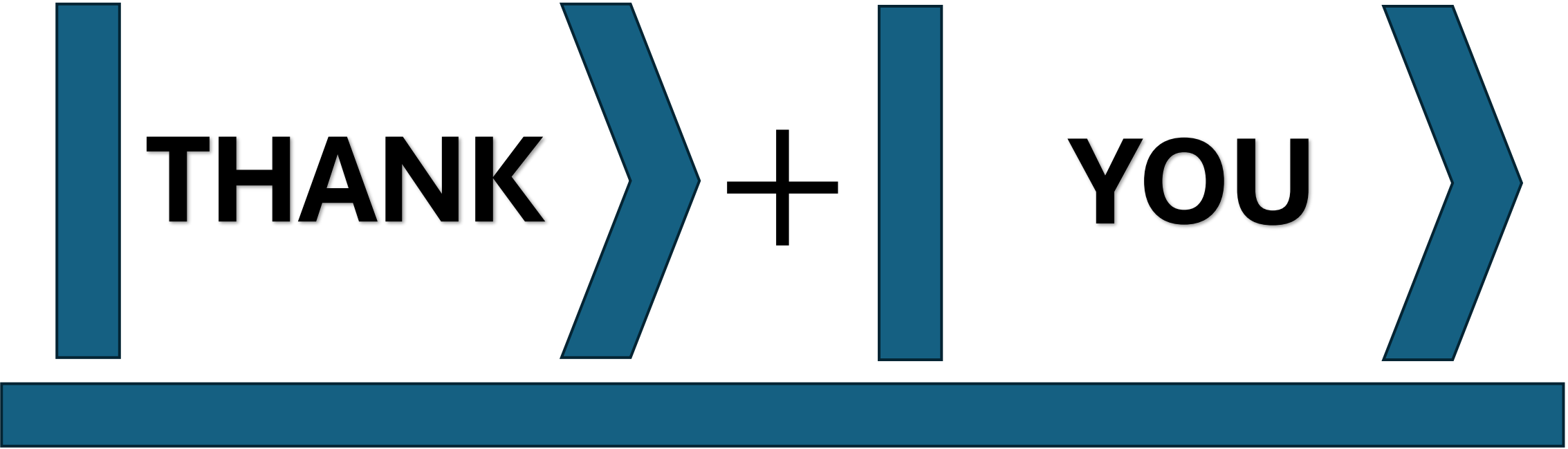
how good is this approximation?

# CONCLUSIONS

Build circuits with a special architecture:

- Find the minimum cut, to estimate  $S_n$  over each circuit.
- Compute the entropies computationally.
- Compare the results and draw conclusions.





The diagram illustrates a quantum state superposition. It features a horizontal blue bar at the bottom. Above this bar, the expression  $\frac{1}{\sqrt{2}}(|\text{THANK}\rangle + |\text{YOU}\rangle)$  is displayed. The words "THANK" and "YOU" are in bold black capital letters. Each word is preceded by a vertical blue bar and followed by a blue chevron pointing to the right. A plus sign is placed between the two chevrons. The entire expression is enclosed in large, light blue parentheses.

**THANK**

**+**

**YOU**

$\sqrt{2}$